
AstroJEPa: Cross-Modal Joint-Embedding Learning for Astronomical Images and Spectra

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Abstract

1 Images and spectra record complementary views of the same galaxy, yet most as-
2 tronomical representation models either learn each modality separately or align
3 them with contrastive negatives. We introduce AstroJEPa, a teacher-free joint-
4 embedding framework that matches augmented image and spectrum views while
5 Sketched Isotropic Gaussian Regularization (SIGReg) prevents collapse. We
6 ask how AstroJEPa should be trained: align frozen unimodal encoders through
7 learned poolers, or learn both encoders jointly from random initialization. On
8 307,428 DESI–Legacy Survey pairs, joint training improves every tested physical-
9 property probe in the shared space, raises cross-modal CKA from 0.720 to 0.770
10 and mean bidirectional recall@10 from 0.87% to 2.97%, and more than doubles
11 shared effective rank. Raw joint backbones retain modality-specific information,
12 including image morphology that the shared projection filters out. A matched
13 contrastive baseline retrieves exact pairs better but is less globally transformable,
14 showing that AstroJEPa’s principal value is a useful, non-collapsed shared scien-
15 tific representation rather than instance matching alone.

16 1 Introduction

17 Modern sky surveys observe the same object through complementary sensors. Galaxy images en-
18 code projected light and morphology; spectra expose redshift, stellar populations, and emission
19 lines. A representation that connects these channels could support cross-survey search, cross-modal
20 imputation, and physical inference with fewer labels. Existing astronomy foundation models have
21 demonstrated the value of large-scale unimodal pretraining and contrastive image–spectrum align-
22 ment [Parker et al., 2024, 2025]. The open question is whether explicit negative pairs are necessary
23 to learn a scientifically useful joint space.

24 Joint-embedding predictive architectures (JEPAs) learn invariances in latent space rather than re-
25 constructing pixels [Assran et al., 2023]. LeJEPa makes this recipe teacher-free by combining an
26 invariance term with SIGReg, which regularizes random one-dimensional projections toward an
27 isotropic Gaussian [Balestrieri and LeCun, 2025]. We adapt that principle across modalities. Astro-
28 JEPa maps two augmented views of a galaxy image and two masked/noisy views of its spectrum to
29 a 256-dimensional shared space, matches all four cross-modal view pairs, and applies SIGReg sepa-
30 rately to each branch. It uses no negatives, queue, EMA teacher, stop-gradient target, reconstruction,
31 or within-modality loss.

32 Our central experiment compares two practical ways to obtain AstroJEPa (Figure 1): (i) indepen-
33 dently pretrain the image and spectrum encoders, freeze them, and align learned-query poolers; or
34 (ii) train both encoders and projectors jointly from scratch. This is deliberately a *strategy com-*
35 *parison*, not an initialization-only ablation: the completed systems differ in trainable capacity and

AstroJEPA learns a shared image--spectrum space while retaining modality-specific backbone states

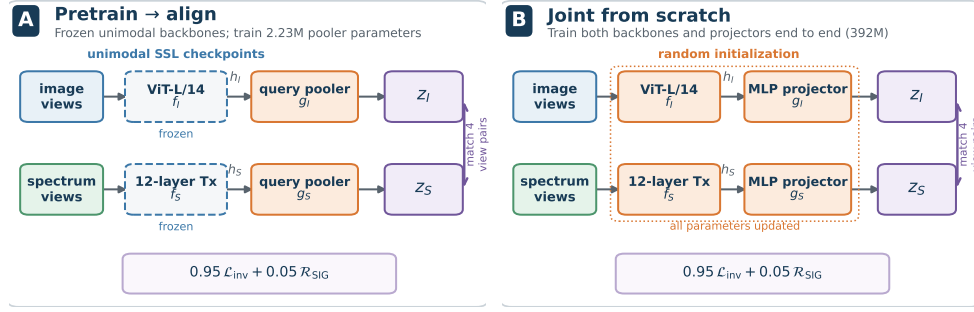


Figure 1: **AstroJEPA and the two training strategies.** Both routes expose raw modality states h_I, h_S and learn shared states z_I, z_S with the same cross-modal invariance–SIGReg objective. Pretrain→Align freezes independently trained backbones and updates two learned-query poolers; Joint-Scratch learns encoders and MLP projectors end to end. Dashed blue boxes are frozen and orange boxes are trainable. Full architectures and objectives are in Appendix A.

duration. We then ask what alignment preserves or removes, and use a controlled contrastive run only as a secondary objective baseline.

The paper makes three contributions:

- **A cross-modal JEPA framework for astronomy.** We specify and validate a non-collapsed, end-to-end image–spectrum model on 307k pairs.
- **A training-strategy study.** Joint learning consistently outperforms the implemented frozen-pooler route in scientific probes, retrieval, cross-modal geometry, and effective rank.
- **A representation audit.** Raw backbones retain useful modality-private information that shared projections intentionally filter; detailed CLIP comparisons and all supporting diagnostics are reported in Appendices E–I.

2 AstroJEPA

Encoders and objective. The image branch is a ViT-L/14 with 1,024-dimensional raw output; the spectrum branch is a 12-layer, 768-dimensional transformer over 389 non-overlapping 20-pixel patches. Modality-specific heads map to $z_I, z_S \in \mathbb{R}^{256}$. For two image views $x_I^{(v)}$ and two spectrum views $x_S^{(u)}$,

$$\mathcal{L}_{\text{AJ}} = 0.95 \frac{1}{4} \sum_{v=1}^2 \sum_{u=1}^2 \left\| z_I^{(v)} - z_S^{(u)} \right\|_2^2 + 0.05 \frac{\mathcal{R}_{\text{SIG}}(z_I) + \mathcal{R}_{\text{SIG}}(z_S)}{2}. \quad (1)$$

SIGReg uses 1,024 random slices and 17 evaluation points. The unnormalized embeddings enter both terms; normalization is used only for cosine evaluation. Although motivated by the JEPA family, the implemented cross-modal loss is direct latent invariance rather than masked predictor-based I-JEPA.

Two training strategies. Pretrain→Align starts from an image ViT-L pretrained on 8.47M galaxies and a spectrum transformer pretrained on 1.10M DESI spectra. Both backbones remain frozen; AstroCLIP-style learned-query attention poolers contribute 2.23M trainable parameters and are optimized for 10 epochs (24k updates). Joint-Scratch randomly initializes both branches and updates the full 392M-parameter system for 50 epochs (60k updates). Both see the same 307,428 paired objects and use AdamW at 10^{-4} , but Pretrain→Align uses global batch 128 whereas Joint-Scratch uses 256. Accordingly, our conclusion is restricted to the two completed strategies; a matched full-finetuning control remains necessary.

63 3 Experiments

64 **Data and evaluation.** The 307,428-pair corpus comprises 197,976 AstroCLIP-mirror objects,
 65 95,895 DESI–Legacy DR10 matches, and 13,557 new DESI–DR8 matches [DESI Collaboration,
 66 2024, Dey et al., 2019]. We evaluate on the 168,280-object mirror with its historical 138,583/29,697
 67 train/test split. Physical probes use the 89,334-object PROVABGS subset [Hahn et al., 2023]; re-
 68 trieval uses every one of the 29,697 test candidates. Because representation training consumed all
 69 307k unlabeled pairs, including mirror test identities, the evaluation is transductive: probe heads and
 70 maps are held out, but the objects are not. Appendices B–C give preprocessing, schedules, hardware,
 71 and split details.

72 3.1 AstroJEPA learns scientifically useful shared representations

Table 1: **Frozen ridge-probe R^2 on shared 256-d AstroJEPA representations.** Joint training improves all five properties for both modalities. Values are means over three probe seeds; standard deviations are below 8×10^{-4} .

Training	View	Redshift	Mass	Metal.	Age	sSFR
Pretrain→Align	Image	.512	.672	.401	.210	.498
	Spectrum	.574	.715	.427	.254	.533
Joint-Scratch	Image	.613	.768	.459	.278	.564
	Spectrum	.632	.780	.448	.274	.570

73 Table 1 establishes the first requirement of a shared scientific space: both modalities linearly expose
 74 redshift and four stellar-population properties. The spectrum branch is strongest, as expected from
 75 the measurement channel, but image embeddings also encode every target. Joint-Scratch improves
 76 mean probe R^2 from 0.459 to 0.537 for images and from 0.500 to 0.541 for spectra. Its raw back-
 77 bones are stronger still (mean 0.562 image and 0.614 spectrum), showing that the shared head is a
 78 selective interface rather than a replacement for modality-specific features.

79 3.2 Joint learning outperforms frozen-pooler alignment

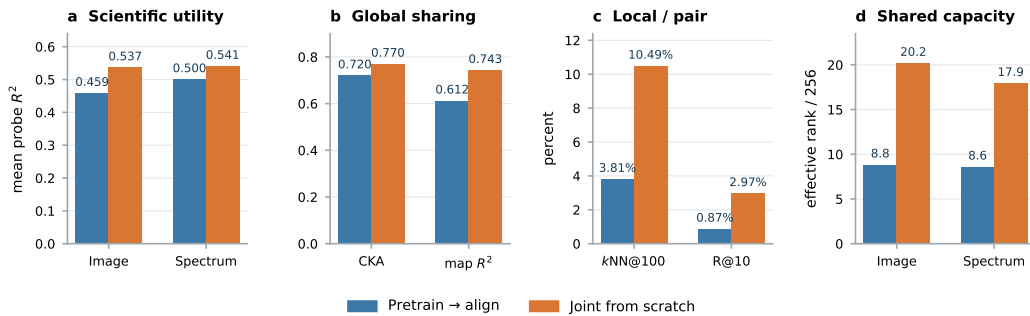


Figure 2: **The implemented joint strategy wins across complementary criteria.** Scientific utility averages the five shared-space probes in Table 1; map R^2 and recall@10 are bidirectional means. CKA uses all 168,280 objects, while neighborhood overlap and retrieval use the 29,697-object test pool. Effective rank is computed from the 256-dimensional shared states.

80 The advantage is not confined to downstream labels (Figure 2). Joint-Scratch raises cross-modal
 81 linear CKA from 0.720 to 0.770 and bidirectional ridge-map R^2 from 0.612 to 0.743. Exact cross-
 82 modal neighborhood overlap at $k = 100$ increases from 3.81% to 10.49%, and bidirectional re-
 83 call@10 from 0.87% to 2.97%. The sequential shared states occupy a narrow manifold—effective
 84 rank 8.83/8.60 for image/spectrum—whereas the joint states reach 20.19/17.94. Thus the end-to-
 85 end route is simultaneously more useful, more mutually predictable, and less dimensionally concen-
 86 trated. These consistent outcomes motivate joint training as our default AstroJEPA recipe, while the
 87 capacity and optimization confound prevents claiming that unimodal pretraining itself is harmful.

88 3.3 What changes during cross-modal learning?

89 The raw joint encoders are not simple copies of the independent ones: same-modality CKA is 0.451
90 for images and 0.398 for spectra; neighborhood overlap@100 is 14.3% and 7.5%. Yet the image
91 backbone retains morphology. On Galaxy10 [Leung and Bovy, 2024], its frozen linear probe reaches
92 $72.22 \pm 1.26\%$ accuracy and $70.19 \pm 1.30\%$ macro-F1, matching the independent image encoder’s
93 $71.83 \pm 0.84\%$ and $70.23 \pm 0.90\%$. The joint shared image projection falls to $44.99 \pm 0.97\%$ accu-
94 racy. The common bottleneck therefore filters image-private morphology even while the backbone
95 preserves it.

96 Cross-modal ridge prediction provides a complementary shared/private audit. For the joint model,
97 the image-predictable part of the spectrum has mean probe R^2 0.531 versus 0.535 for the total
98 spectrum state; its residual still scores 0.107. The spectrum-predictable image component scores
99 0.534 versus 0.531 total, with residual 0.071. This is a linear decomposition, not mutual information:
100 residuals also contain nonlinear shared structure and mapping error. The result nevertheless supports
101 exposing both (z_m, h_m) rather than only a single common vector.

102 3.4 A compact contrastive control

103 We also train CLIP from random initialization for the same 50 epochs, paired objects, architectures,
104 and views [Radford et al., 2021]. CLIP improves bidirectional recall@10 from 2.97% to 7.98%,
105 but AstroJEPA has higher projected CKA (0.770 vs. 0.634), pairwise-geometry correlation (0.607
106 vs. 0.485), and bidirectional linear-map R^2 (0.749/0.736 vs. 0.551/0.549). This result bounds the
107 claim: AstroJEPA does not dominate exact instance matching; it learns a more globally congruent
108 and linearly transformable shared space under these runs. The full comparison—raw and projected
109 probes, CCA, Procrustes retrieval, decoder transfer, eigenspectra, and all plots—is intentionally
110 moved to Appendix G.

111 4 Discussion

112 A frozen pooler must discover the intersection of two independently organized token spaces, whereas
113 paired end-to-end gradients can reorganize both encoders. The sequential ranks near nine are con-
114 sistent with a narrow intersection but do not prove this mechanism because head and budget also
115 change. Contrastive negatives sharpen instance ranking, while AstroJEPA’s invariance makes modal-
116 ities more linearly transformable; neither property alone captures quality. For science, the shared
117 state can support retrieval and cross-modal prediction, while the raw state retains channel-specific
118 morphology or spectral detail. Exposing both makes information discarded by alignment measur-
119 able.

120 5 Limitations and impact

121 Each principal representation has one training seed, so deterministic full-pool diagnostics do not
122 quantify training-seed variance. The strategies also differ in parameters, duration, batch, and head
123 type. Evaluation is transductive and DESI-dominated; retrieval counts valid analogues and dupli-
124 cates as errors. A definitive test needs object-disjoint pretraining, matched full-model fine-tuning,
125 multiple seeds, and a no-SIGReg control.

126 AstroJEPA may reduce labeling needs and enable large-catalog search or cross-modal imputation,
127 but it inherits survey selection, calibration, cross-match, and physical-label biases. Embeddings are
128 not calibrated physical estimators and require downstream validation. The data are public astro-
129 nomical observations and contain no personal data; the main concern is compute and the risk of
130 overconfident scientific use.

131 **Conclusion.** AstroJEPA learns useful shared image–spectrum representations without contrastive
132 negatives. For the completed systems, joint training is strongest, while raw states preserve science
133 filtered by the common bottleneck.

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A Full AstroJEPA Architecture and Objectives

A.1 Terminology and scope

We use *AstroJEPA* for the complete image–spectrum representation framework, and *JEPA+SIGReg* when naming its optimization objective in control tables. The system follows the joint-embedding principle—predict or match targets in representation space rather than reconstructing inputs—and uses LeJEPA’s teacher-free distributional regularization. It is not the canonical masked-image I-JEPA architecture: there is no context encoder, EMA target encoder, predictor, or stop-gradient branch. This distinction matters because Equation 1 is direct cross-modal invariance.

A.2 Backbones and heads

Table 2: Architecture used by the completed cross-modal runs.

Branch	Encoder	Raw dim.	Tokens	Shared dim.
Image	ViT-L/14, 24 blocks, 16 heads	1,024	$10 \times 10 + 1$	256
Spectrum	Transformer, 12 blocks, 12 heads, MLP ratio 4	768	$389 + 1$	256

The image encoder consumes 140×140 three-channel astronomy images. The spectrum encoder consumes a fixed 7,781-pixel DESI grid. It retains the first 7,780 pixels as 389 disjoint patches of width 20 and adds a CLS token. A patch is attendable when at least half of its pixels are scientifically valid; invalid padding, physical masks, and artificial augmentation masks remain distinct.

In Joint-Scratch, each CLS state enters a modality-specific MLP projector. In Pretrain→Align, token sequences enter independent learned-query cross-attention poolers with four attention heads, dropout 0.1, a residual connection, and a fourfold-width MLP; the frozen image and spectrum tensors are not copied into the alignment checkpoint. Total parameters are 391,990,528, of which 2,234,368 are trainable in Pretrain→Align and approximately 392M in Joint-Scratch.

A.3 Cross-modal AstroJEPA loss

Let B be the local batch and $V_I = V_S = 2$. For each object, Equation 1 averages all $V_I V_S = 4$ squared-error pairs. SIGReg applies independent random projections to the image and spectrum embedding batches:

$$\mathcal{R}_{\text{SIG}} = \frac{1}{K} \sum_{k=1}^K \text{EP}(\{\langle z_b, \theta_k \rangle\}_{b=1}^B, \mathcal{N}(0, 1)), \quad K = 1024,$$

where EP is the Epps–Pulley normality statistic evaluated at 17 points. Random slice seeds and differentiable distributed moments are synchronized across ranks. The regularizer sees the effective distributed batch without requiring a full embedding all-gather. No image–image, spectrum–spectrum, local-patch, reconstruction, contrastive, or distillation term is active in the completed cross-modal runs.

A.4 Independent unimodal objectives

The image SSL model receives two 140-pixel global crops and eight 56-pixel local crops. Its invariance target is the mean of the two global projected states and its loss is $0.95 \mathcal{L}_{\text{view}} + 0.05 \mathcal{R}_{\text{SIG}}$. The 1,024-dimensional ViT-L state is followed during training by a 1,024–2,048–2,048–64 projector; downstream evaluation normally uses the raw backbone.

Spectrum SSL receives two independently masked and noise-perturbed views and optimizes

$$\mathcal{L}_{\text{spec}} = \underbrace{0.95 \mathcal{L}_{\text{CLS}} + 0.05 \mathcal{R}_{\text{SIG}}^{\text{global}}}_{\mathcal{L}_{\text{global}}} + \underbrace{0.95 \mathcal{L}_{\text{patch}} + 0.05 \mathcal{R}_{\text{SIG}}^{\text{local}}}_{\mathcal{L}_{\text{local}}}.$$

Patch comparisons use the same wavelength positions and exclude invalid targets. The 768-dimensional raw state and 64-dimensional objective head are both evaluated in Appendix D.

A.5 Contrastive control

The secondary scratch CLIP control uses the same encoders, projectors, paired examples, and four view combinations as Joint-Scratch. It ℓ_2 -normalizes projected embeddings and averages symmetric image-to-spectrum and spectrum-to-image InfoNCE. Differentiable all-gather supplies every object in the global batch as a candidate. Temperature is learned from 0.07 and the multiplicative logit scale is capped at 100. SIGReg and same-modality terms are absent. A separate frozen-backbone control uses the same learned-query topology as Pretrain→Align, a 512-dimensional output, fixed logit scale 15.5, and global-batch InfoNCE; its results are isolated in Appendix G.

B Data and Preprocessing

B.1 Corpus inventory

Table 3: Datasets used for representation training and evaluation. Counts are objects after the stated local filtering; Galaxy10 DECaLS is from Leung and Bovy [2024].

Dataset	Objects	Role
Smith42/galaxies	8,474,566	streamed image SSL
DESI EDR/SV3 spectra	1,126,441	spectrum SSL; 1,103,874 train
Paired DESI–Legacy union	307,428	cross-modal representation training
AstroCLIP benchmark mirror	168,280	maps, retrieval, redshift probes
PROVABGS valid subset	89,334	four stellar-population probes
Galaxy10 DECaLS	17,736	10-way morphology

The spectrum SSL object-ID split contains 1,103,874 train, 11,280 validation, and 11,287 test spectra. The evaluation mirror retains its historical 138,583/29,697 split. After requiring valid PROVABGS labels, 73,341 objects remain in train and 15,993 in test. Galaxy10 uses three stratified 80/10/10 splits; checkpoint selection uses validation macro-F1.

Table 4: Composition and harmonization of the 307,428-pair corpus.

Source	Rows	Fraction	Model-facing contract
AstroCLIP mirror	197,976	64.40%	flux-only DESI; raw $g/r/z$ nanomaggies
MMU DESI–DR10	95,895	31.19%	flux/ivar/mask; rendered RGB PNG
Manual DESI–DR8	13,557	4.41%	flux/ivar/mask; DR8 crop
Total	307,428	100%	fixed spectrum grid; two 140-pixel image views

The manual cross-match initially contained 13,579 new target IDs; 21 missing image IDs removed 22 pair rows during materialization. No spectrum payload was missing. AstroCLIP nanomaggy images use the published $g/r/z$ arcsinh RGB mapping before common augmentations. Every spectrum is mapped to the same 7,781-pixel grid and normalized by its valid-pixel mean and standard deviation. MMU/manual rows retain pipeline inverse variance and masks; flux-only AstroCLIP rows mark finite values valid and disable uncertainty noise rather than synthesizing inverse variance.

B.2 Augmentations

Images use random resized crops, horizontal/vertical flips, rotations, and astronomy-specific blur/noise transforms. Independent image SSL uses two 140-pixel global and eight 56-pixel local views; cross-modal training uses two 140-pixel global views. Spectra use two views with 30% disjoint span masks, span length 2–12 patches, and inverse-variance-scaled Gaussian noise with scale 0.5 capped at normalized standard deviation 3. Positive inverse variance determines augmentation noise but is not an encoder channel or loss weight. Wavelength is implicit in fixed token position; wavelength and line-spread function are validated metadata, and LSF is not yet an input.

B.3 Evaluation leakage boundary

Cross-modal SSL intentionally uses all available pairs without historical split labels. Consequently, representations may have seen the unlabeled identity of objects later used in the mirror test set. All ridge probes, cross-modal maps, and hyperparameter selection remain train-only, but this does not make the backbone test object-disjoint. We therefore use these results for controlled representation comparison and do not present them as strict unseen-object generalization.

C Training and Evaluation Details

Table 5: Completed principal training runs. Samples counts source-object presentations, not augmented views.

Run	Trainable	GPUs	Global batch	Epochs	Updates	Samples
Image SSL ViT-L	310.8M	4	768	5 passes	54,940	42.19M
Spectrum SSL v2	87M	4	64	10	172,390	11.04M
Pretrain→Align	2.23M	4	128	10	24,000	3.07M
Frozen InfoNCE	7.09M	4	256	10	12,000	3.07M
Joint-Scratch	~392M	4	256	50	60,000	15.36M
CLIP scratch	~392M	4	128	50	120,000	15.36M

All four cross-modal configurations use AdamW with learning rate 10^{-4} , weight decay 0.05, 1,000 warmup updates, cosine decay to 10^{-6} , bfloat16, gradient-norm clipping at 1, and seed 42. Joint-Scratch ran for 10 h 13 min and CLIP for 11 h 5 min on four GPUs. They match architectures, views, epochs, sample exposure, and approximately wall time, but not global batch or update granularity. The frozen Pretrain→Align run uses the same optimizer family and paired corpus but is intentionally shorter.

Image SSL uses AdamW at 5×10^{-4} , weight decay 0.05, 1,000-step warmup, cosine decay to 10^{-6} , and bfloat16. The five streamed passes correspond to 42.19M source-image presentations and 421.94M augmented views. Spectrum v2 uses the same optimizer values and numerical settings for ten true passes. Its corrected schedule is 17,239 synchronized updates per epoch.

Frozen probes. Continuous targets are standardized from the training set. Ridge regularization is selected on a train-only validation carve-out and the model is refit on the remaining training rows. Reported values average three deterministic split seeds. Galaxy10 uses a class-balanced linear classifier; accuracy and macro-F1 are means and standard deviations over three stratified splits.

Alignment diagnostics. Exact retrieval ranks every candidate by cosine similarity. Cross-modal k NN overlap is $|N_k^I(x) \cap N_k^S(x)|/k$, averaged over queries. Linear CKA uses centered Gram matrices. Pairwise geometry samples 500,000 deterministic object pairs and correlates within-modality cosine similarities with Spearman’s ρ . Cross-modal ridge maps select their penalty on training data and report held-out global R^2 . CCA, orthogonal Procrustes, covariance eigenspectra, participation ratio, and effective rank provide complementary subspace diagnostics.

D Independent Unimodal SSL

D.1 Image SSL

Table 6: Controlled Galaxy10 frozen linear probes.

Backbone	Selected update	Accuracy (%)	Macro-F1 (%)
ResNet9	8,000	44.80 ± 1.75	42.69 ± 1.70
ViT-S/14	21,000	61.19 ± 1.42	59.34 ± 1.16
ViT-L/14	52,000	71.83 ± 0.84	70.23 ± 0.90

The ViT-L has 304.37M backbone parameters and 6.43M projector parameters. A two-layer MLP probe reached approximately 72.4% accuracy, only a small gain over the linear head. The recorded

263 DINOv2 ViT-g/14 and AION-1-L contexts are 71.4% and 87.2%, respectively [Oquab et al., 2024,
 264 Parker et al., 2025]; these use different pretraining, data, and downstream heads and are not a
 265 matched leaderboard.

Table 7: Fixed-seed image-redshift R^2 over the ViT-L checkpoint series. The broad late plateau motivated selecting update 52k by Galaxy10 validation macro-F1 rather than redshift test performance.

Update (k)	4	8	12	16	20	24	28	32
R^2	.51761	.52786	.52551	.52723	.52938	.53032	.53033	.53018
Update (k)	36	40	44	48	52	54.94	CPT 62.99	AstroDINO
R^2	.53117	.53167	.53137	.53089	.53028	.53023	.53128	.52888

266 The one-epoch local-data continuation improves its 52k source by only about 0.001 R^2 . The historical
 267 Galaxy10 redshift probe reports 0.748 after restricting to $z < 0.25$; it is a different sample and
 268 is not compared numerically with the DESI–Legacy mirror.

269 D.2 Spectrum SSL

Table 8: Frozen spectrum probe R^2 . V2 is the corrected global+local LeJEPa run; projected is its 64-dimensional objective head.

Representation	Redshift	Mass	Metal.	Age	sSFR
Spectrum v1 raw	.432	.506	.235	.185	.510
Spectrum v2 raw	.555	.698	.409	.232	.496
Spectrum v2 projected	.388	.541	.259	.104	.430

270 V2 improves four of five raw targets substantially but slightly regresses sSFR. Its raw effective rank
 271 is 13.67/768 (participation ratio 9.59, largest eigenvalue share 0.208); the 64-dimensional projector
 272 has effective rank 23.03 (participation ratio 17.36, largest share 0.117). The lower projected probe
 273 scores reinforce the practice of using the backbone, not the SSL head, for unimodal science.

Table 9: Linear and two-layer MLP probes on the Joint-Scratch raw spectrum state. The nonlinear head principally helps redshift and age.

Probe	Redshift	Mass	Metal.	Age	sSFR
Ridge	.676	.843	.549	.366	.639
Two-layer MLP	.702 $\pm$.003	.854	.580	.410 $\pm$.004	.668

274 The earlier 95k-pair scratch pilot reached raw spectrum R^2 0.642/0.823/0.533/0.321/0.629 (red-
 275 shift/mass/metallicity/age/sSFR), but its loader under-consumed rows per nominal epoch. Its image
 276 redshift scores were .523 raw and .517 shared; raw effective rank was 9.64/7.04 for image/spectrum
 277 and shared rank was 8.58/10.07. Its spectrum shared scores were .588/.775/.443/.249/.588 in the
 278 same target order. It is retained as 8,396-update feasibility evidence, not as a controlled 50-epoch
 279 result.

280 The dedicated unimodal battery and the later strategy-study cache give slightly different values after
 281 their respective evaluation pipelines (for example, spectrum-v2 mass .698 here and .695 in Table 10).
 282 All within-strategy conclusions use one pipeline consistently.

E Cross-Modal Training: Pretrained vs. Scratch

E.1 Complete downstream probes

Table 10: All frozen ridge-probe R^2 for the two AstroJEPa strategies. Raw is the backbone state; shared is the objective-facing 256-d state.

Training	Modality	State	Redshift	Mass	Metal.	Age	sSFR
Pretrain→Align	Image	raw	.530	.691	.413	.241	.501
	Image	shared	.512	.672	.401	.210	.498
	Spectrum	raw	.555	.695	.409	.228	.496
	Spectrum	shared	.574	.715	.427	.254	.533
Joint-Scratch	Image	raw	.627	.806	.474	.314	.591
	Image	shared	.613	.768	.459	.278	.564
	Spectrum	raw	.675	.842	.549	.363	.639
	Spectrum	shared	.632	.780	.448	.274	.570

The exact CSV means are (.511772, .672491, .400697, .210183, .497528) for sequential image shared; (.573527, .714941, .427080, .253727, .532742) for sequential spectrum shared; (.613104, .768012, .459308, .278290, .563879) for joint image shared; and (.632093, .779902, .448309, .273529, .569867) for joint spectrum shared. Displayed rounding does not change any comparison.

E.2 Retrieval and shared geometry

Table 11: Exact retrieval over the 29,697-object test pool. Recall is percent.

Training	Direction	R@1	R@5	R@10	Median rank	MRR
Pretrain→Align	I→S	.074	.404	.849	1,432	.00535
	S→I	.091	.505	.892	1,404	.00564
Joint-Scratch	I→S	.461	1.805	3.307	429	.01786
	S→I	.320	1.431	2.633	454	.01475

On the complete 168,280-object pool, bidirectional mean R@10 is 0.199% for Pretrain→Align and 0.626% for Joint-Scratch.

Table 12: Complete shared-space geometry and capacity summary. Map values are held-out global R^2 ; CCA values are mean canonical correlation.

Metric	Pretrain→Align	Joint-Scratch
Cross-modal linear CKA	.720	.770
k NN overlap@100	3.81%	10.49%
Ridge map I→S	.598	.749
Ridge map S→I	.627	.736
CCA top 10	.728	.916
CCA top 50	.310	.675
Effective rank, image shared	8.83	20.19
Effective rank, spectrum shared	8.60	17.94

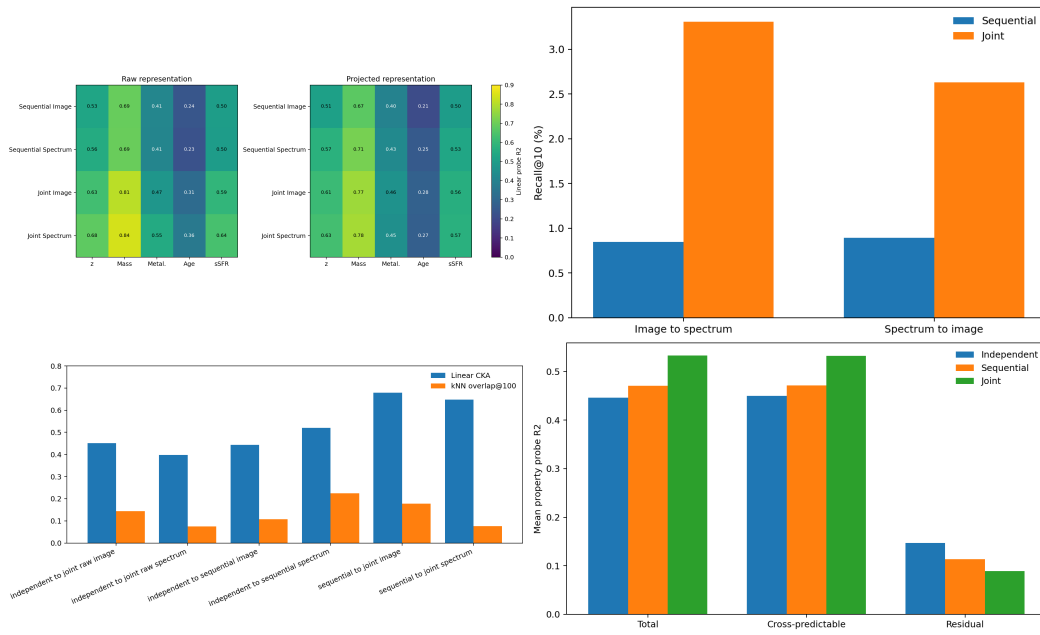


Figure 3: The complete strategy-comparison figure set: downstream accessibility, exact pair retrieval, representation retention, and shared/private accessibility. Numeric values appear in Tables 10–13.

F Representation Analysis Before and After Alignment

F.1 Raw representation retention

Table 13: Same-modality retention between independently pretrained, sequential, and joint raw states. Spearman is sampled pairwise-cosine correlation.

Comparison	Modality	CKA	Spearman	kNN@10	kNN@50	kNN@100
Independent→Joint	Image	.451	.347	8.42%	12.18%	14.33%
	Spectrum	.398	.280	2.26%	5.30%	7.50%
Independent→Sequential	Image	.443	.356	5.17%	8.47%	10.78%
	Spectrum	.520	.503	12.60%	18.94%	22.46%
Sequential→Joint	Image	.678	.433	4.09%	11.68%	17.76%
	Spectrum	.648	.459	1.57%	4.73%	7.63%

The sequential raw tensors are exactly the independent backbones, but the sequential aligned pooler summarizes token sequences rather than simply using the standalone CLS vector. For that reason, comparisons involving sequential outputs are not identities even though the source backbone weights are frozen.

F.2 Linear shared/private accessibility

For a target state Y and the opposite modality X , fit a train-only ridge map $\hat{Y} = AX$. We call $\hat{Y}$ shared and $Y - \hat{Y}$ private, then fit the same scientific probes separately. The construction is operational: it does not estimate mutual information or guarantee statistical independence.

Table 14: Mean R^2 across redshift, mass, metallicity, age, and sSFR for total, cross-modally predictable, and residual components. Map is the global held-out cross-modal R^2 .

State	Predicted target	Total	Shared	Residual	Map
Independent	Spectrum	.455	.440	.156	.025
	Image	.438	.459	.138	.025
Pretrain→Align	Spectrum	.493	.452	.152	.597
	Image	.449	.491	.075	.626
Joint-Scratch	Spectrum	.535	.531	.107	.749
	Image	.531	.534	.071	.736

Shared-component scores can slightly exceed total scores because the learned cross-modal map suppresses probe-irrelevant variation. Residual scores need not vanish because nonlinear shared signal, finite-sample mapping error, and modality-private information are all present.

F.3 Morphology retention

Table 15: Galaxy10 morphology retained by image states.

Image state	Accuracy (%)	Macro-F1 (%)
Independent raw	71.83 ± 0.84	70.23 ± 0.90
Pretrain→Align shared	59.63 ± 1.20	57.53 ± 1.14
Joint-Scratch raw	72.22 ± 1.26	70.19 ± 1.30
Joint-Scratch shared	44.99 ± 0.97	42.77 ± 0.97

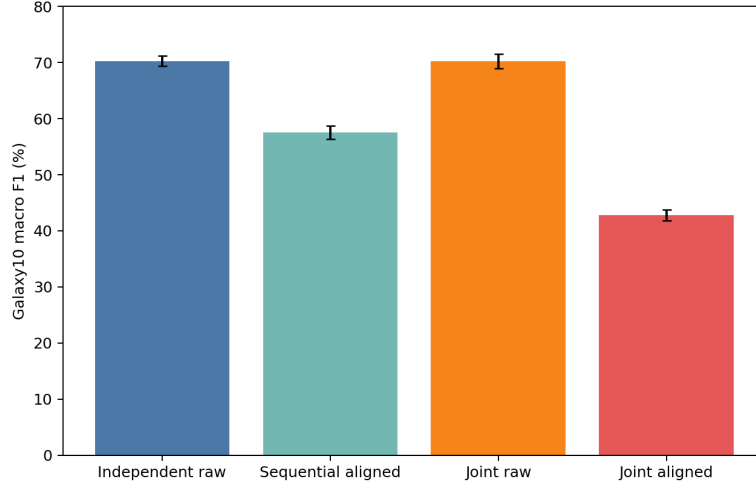


Figure 4: The joint raw backbone preserves morphology while the shared projection filters it. Error bars are standard deviations across three stratified Galaxy10 splits.

G Contrastive Baselines: Detailed JEPA vs. CLIP

G.1 Frozen-backbone objective ablation

Before the end-to-end comparison, we aligned the same frozen unimodal backbones with two learned-query adapter systems. The JEPA+SIGReg adapter has 256-dimensional output and 2.23M trainable parameters; the released-code-style InfoNCE adapter has 512-dimensional output and 7.09M trainable parameters. Both use the same 307,428 pairs for 10 epochs, but JEPA+SIGReg uses global batch 128 for 24,000 updates and InfoNCE uses global batch 256 for 12,000. The objective, dimensionality, parameter count, and update granularity are therefore jointly varied.

Table 16: Frozen-adapter objective ablation, with the two end-to-end runs shown for context. Values are shared-space probe R^2 .

Target	Post JEPA	Post InfoNCE	Scratch JEPA	Scratch CLIP
Image redshift	.5118	.5417	.6137	.6016
Spectrum redshift	.5739	.5868	.6322	.6515
Stellar mass	.7176	.7482	.7815	.8169
sSFR	.5330	.5664	.5701	.6279
Metallicity	.4282	.4424	.4482	.5089
Stellar age	.2584	.2643	.2758	.3555

InfoNCE improves every tested frozen-adapter target. Its image/spectrum effective ranks are 12.01/12.85 in 512 dimensions, versus 8.83/8.60 in 256 dimensions for JEPA+SIGReg; the unequal dimensions prevent a strict rank comparison. Both end-to-end systems outperform both frozen adapters on every row. This supports paired backbone adaptation, while leaving open whether fine-tuned unimodal initialization would outperform random initialization.

G.2 Control boundary

The scratch JEPA+SIGReg and CLIP runs use identical randomly initialized architectures, paired objects, view generators, and 50-epoch sample exposure. JEPA+SIGReg uses global batch 256 and 60,000 updates; CLIP uses global batch 128, 127 in-batch negatives per anchor, and 120,000 updates. Thus the comparison is architecture- and sample-matched but partially confounds objective, batch, and optimizer granularity.

Table 17: Exact test-pool retrieval. Recall is percent.

Objective	Direction	R@1	R@5	R@10	Median rank	MRR
JEPA+SIGReg	I→S	.461	1.805	3.307	429	.01786
	S→I	.320	1.431	2.633	454	.01475
CLIP	I→S	1.488	5.378	8.772	230	.04235
	S→I	1.054	4.152	7.193	270	.03449

Table 18: Global, local, and information alignment. Neighborhood values are the machine-readable CSV results; they supersede rounded draft-report values.

Metric	JEPA+SIGReg	CLIP
Projected cross-modal CKA	.7705	.6338
Raw cross-modal CKA	.8983	.8320
Pairwise-geometry Spearman	.6073	.4845
k NN overlap@10	1.702%	2.621%
k NN overlap@50	6.263%	7.256%
k NN overlap@100	10.488%	10.969%
Map I→S R^2	.7492	.5507
Map S→I R^2	.7365	.5491
CCA top 10	.9156	.9046
CCA top 50	.6750	.6579
CCA all dimensions	.2118	.2156

325 The two metric families are not contradictory. Exact retrieval is a tail-ranking statistic, whereas
326 CKA, sampled distance correlation, and map R^2 summarize many global relations. CLIP’s negatives
327 directly train competitive instance ranking; SIGReg-supported invariance produces a more linearly
328 transformable manifold.

329 G.3 Scientific probes and dimensional use

Table 19: All raw and projected probe R^2 for the scratch objective comparison. Bold marks the stronger objective within each modality/state.

Objective	Modality	State	Redshift	Mass	Metal.	Age	sSFR
JEPA+SIGReg	Image	raw	.627	.806	.474	.314	.591
	Image	shared	.613	.768	.459	.278	.564
	Spectrum	raw	.675	.842	.549	.363	.639
	Spectrum	shared	.632	.780	.448	.274	.570
CLIP	Image	raw	.610	.793	.465	.307	.580
	Image	shared	.601	.792	.469	.306	.583
	Spectrum	raw	.678	.846	.567	.391	.663
	Spectrum	shared	.651	.816	.509	.353	.628

Table 20: Effective rank. CLIP distributes variance over more directions, while JEPA+SIGReg yields stronger global cross-modal congruence.

Objective	Image raw	Spectrum raw	Image shared	Spectrum shared
JEPA+SIGReg	28.26	26.08	20.19	17.94
CLIP	64.29	58.14	45.39	44.65

330 Same-modality CKA between objectives is 0.9395 for raw image states and 0.8528 for raw spectra.
 331 The backbones therefore remain broadly related even though the shared heads and cross-modal
 332 behavior differ.

333 G.4 Linear alignment and decoder transfer

334 After per-modality train-set standardization, an orthogonal Procrustes map modestly improves exact
 335 retrieval: JEPA+SIGReg $I \rightarrow S$ $R@1$ rises from 0.44% to 0.54% and $S \rightarrow I$ from 0.30% to 0.45%;
 336 CLIP rises from 1.55% to 2.02% and 1.49% to 1.84%, respectively. Matched cosine similarity
 337 changes from 0.780 to 0.806 for JEPA+SIGReg and from 0.402 to 0.595 for CLIP. These stan-
 338 dardized before-values are not the raw-cosine baselines in Table 17. The limited retrieval recovery
 339 despite high CCA implies more than a simple rotation separates the modalities.

Table 21: Probe-decoder transfer after Procrustes alignment. Each decoder is trained on the source named in the direction and evaluated on the other modality. Values are R^2 .

Objective	Decoder transfer	Redshift	Mass	Metal	Age	sSFR
JEPA+SIGReg	Image decoder on spectrum	.515	.621	.369	.166	.474
CLIP	Image decoder on spectrum	.467	.626	.355	.225	.365
JEPA+SIGReg	Spectrum decoder on image	.529	.630	.373	.219	.495
CLIP	Spectrum decoder on image	.502	.666	.372	.214	.441

340 Direct decoder transfer without alignment is frequently poor or negative. The complete own-space,
 341 direct-transfer, and Procrustes-transfer values are visualized in Figure 6 and stored with the study
 342 CSVs.

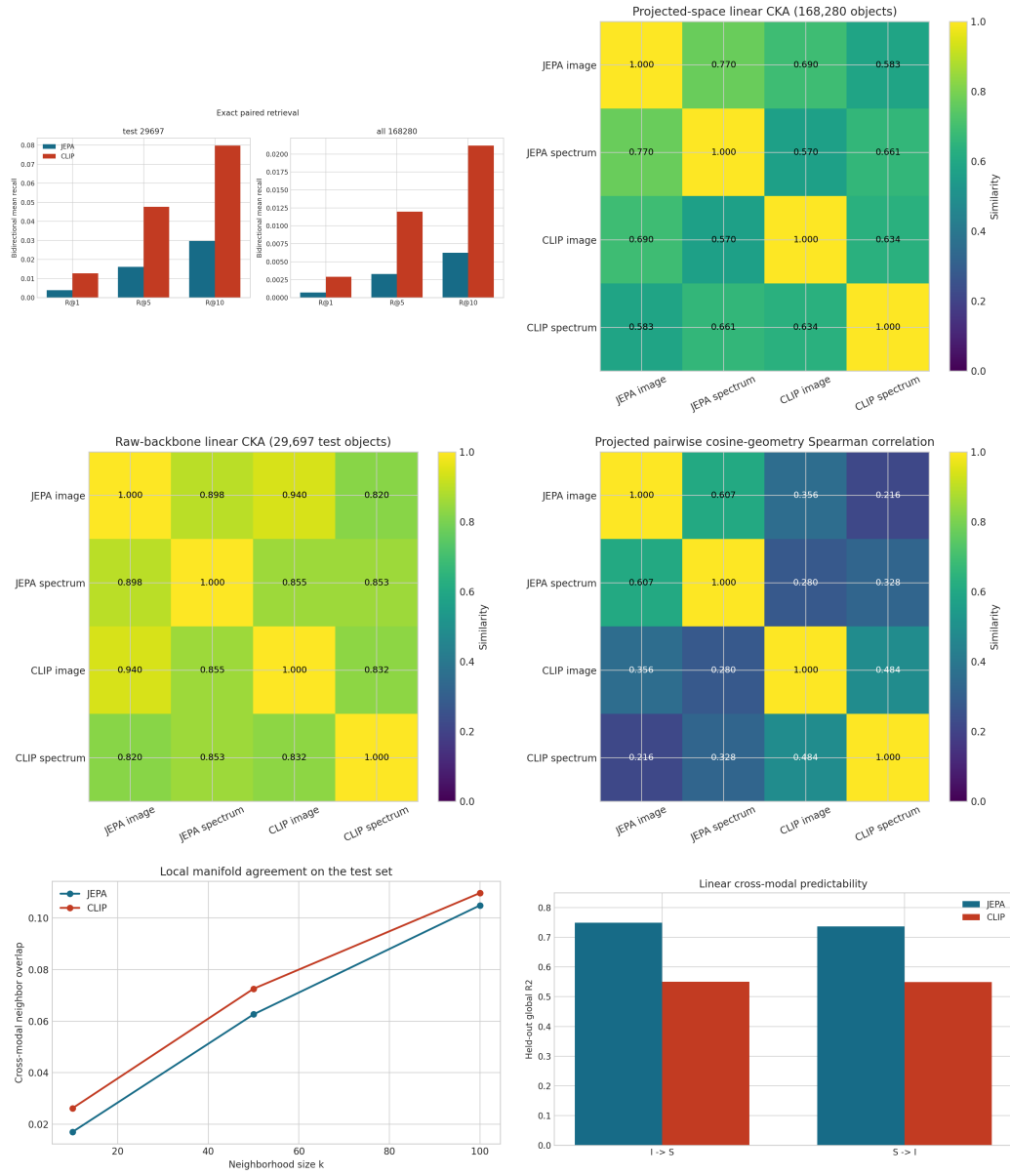


Figure 5: Primary JEPA+SIGReg versus CLIP diagnostics: exact retrieval, projected and raw CKA, pairwise geometry, local-neighborhood overlap, and linear cross-modal predictability.

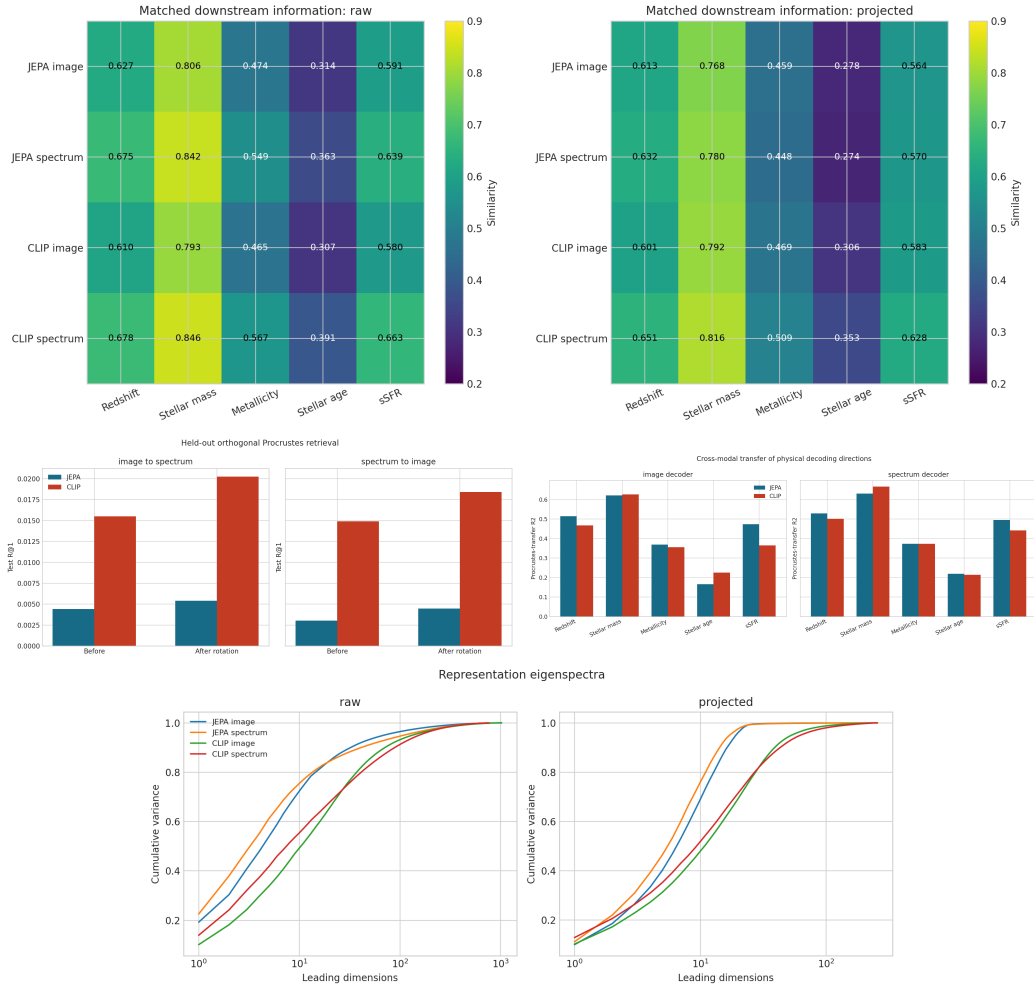


Figure 6: Additional objective diagnostics: raw/projected scientific probes, Procrustes retrieval, decoder transfer (including own, direct, and aligned scores), and covariance eigenspectra.

H Additional Downstream Results

H.1 Galaxy10 per-class behavior

Table 22: Per-class recall and F1 for the selected independent ViT-L/14 checkpoint.

Class	Recall	F1	Class	Recall	F1
Disturbed	.486	.457	Merging	.771	.762
Round smooth	.863	.847	In-between smooth	.786	.794
Cigar smooth	.824	.609	Barred spiral	.628	.643
Tight spiral	.652	.634	Loose spiral	.486	.546
Edge-on, no bulge	.893	.884	Edge-on, bulge	.862	.847

Edge-on and smooth classes are strongest; disturbed and loose-spiral galaxies remain difficult. The model-selection curves and radar visualization appear in Appendix I.

H.2 Galaxy Zoo DECaLS question-wise probes

Table 23: Public-split GZD-5 frozen-probe results for image SSL ViT-L/14 at 52k. F1 is macro-F1 within each question.

Question	Classes	Acc.	F1	Question	Classes	Acc.	F1
Smooth	3	.774	.676	Disk edge-on	2	.885	.830
Spiral arms	2	.935	.946	Bar	3	.538	.376
Bulge size	5	.776	.762	Roundedness	3	.825	.825
Edge-on bulge	3	.800	.711	Spiral winding	3	.752	.646
Arm count	6	.422	.386	Merging	4	.814	.731
Question mean		.752	.689				

The historical AstroCLIP context is 0.761 mean accuracy and 0.743 mean F1, but its published head and evaluation details are not identical. An early local reproduction also reached 0.761 accuracy but contained a majority-class/protocol failure and is not used. A stricter corrected variant measured 0.655 ± 0.013 mean F1. We retain these variants to document protocol sensitivity rather than average them into one headline number.

H.3 Published context and comparability

The independent image ViT-L’s 71.83% Galaxy10 accuracy is close to a 71.4% DINOv2 reference and below AION-1-L’s 87.2%; AION uses an MLP head and a much larger multimodal corpus. Likewise, recorded AstroCLIP spectrum values (redshift .98, mass .88, sSFR .64, metallicity .58, age .43) use a different sample and training history. Our scratch CLIP spectrum approaches several of these physical-property numbers but not redshift. These values provide context only and are not included in matched statistical comparisons.

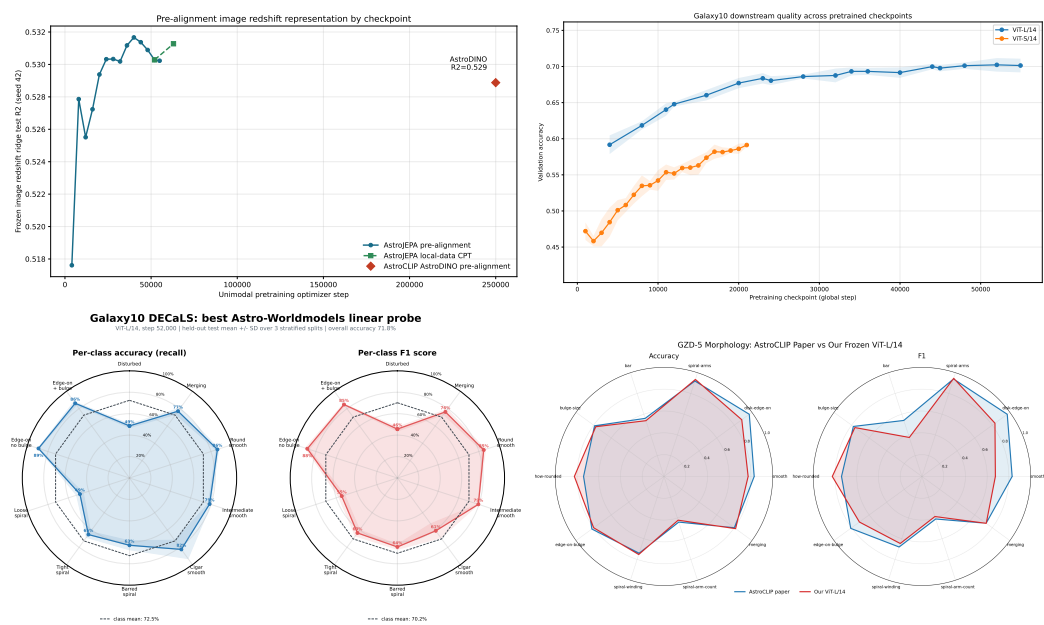


Figure 7: Unimodal image evaluation: fixed-seed redshift progression, Galaxy10 checkpoint selection, selected-model per-class behavior, and public-split GZD-5 context.

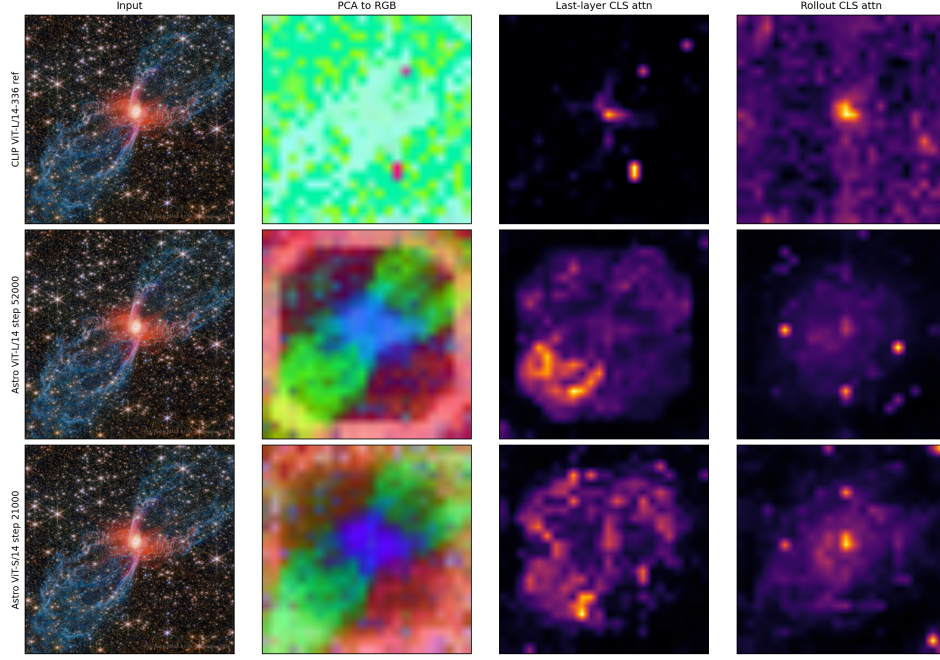


Figure 8: Patch-token PCA and CLS-attention comparison for the selected astronomy ViTs and reference image encoder. These visualizations show structured, object-centered responses but are qualitative diagnostics, not independent evidence for semantic localization.

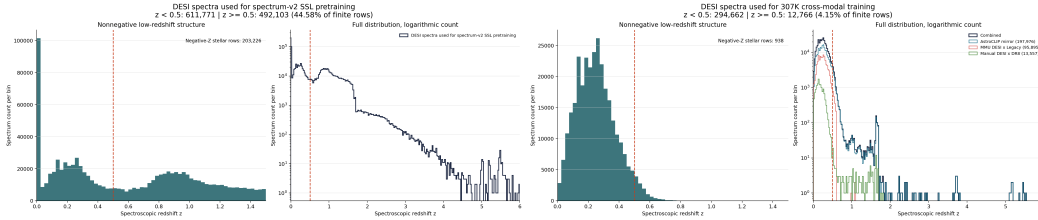


Figure 9: Redshift coverage differs sharply between spectrum SSL and paired cross-modal training. The spectrum SSL train set contains 492,103/1,103,874 (44.58%) objects at $z \geq 0.5$, whereas the paired union contains 12,766/307,428 (4.15%). This shift is a plausible contributor to the cross-modal spectrum's redshift behavior and limits generalization beyond the paired population.

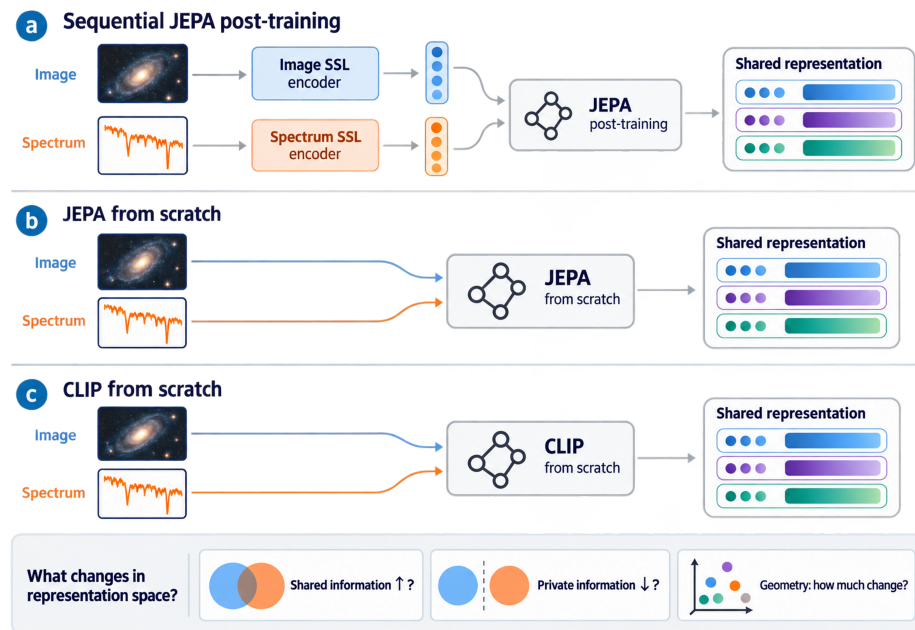


Figure 10: Expanded project schematic retained from the objective-comparison study. It distinguishes sequential AstroJEPA, joint AstroJEPA, and a scratch CLIP control, and motivates the shared/private/geometry diagnostics. Figure 1, rather than this control-oriented diagram, is the primary architecture figure.

J Additional Ablations, Null Results, and Reproducibility Inventory

J.1 Completed negative or qualified results

- **Image continuation pretraining:** one extra local-data epoch changes fixed-seed redshift R^2 from .53028 at the source checkpoint to .53128, insufficient evidence for a new scaling regime.
- **Spectrum v2 is not uniformly better:** it improves redshift, mass, metallicity, and age over v1 but changes sSFR from .510 to .496.
- **Objective heads are selective:** spectrum-v2 projected probes are uniformly below raw probes; joint shared states also trail raw states on most properties and sharply reduce Galaxy10 morphology.
- **Linear versus nonlinear probes:** a two-layer MLP materially improves joint-spectrum redshift and age, while the matched image MLP does not consistently improve on ridge.
- **Procrustes is insufficient:** orthogonal alignment improves retrieval only modestly for both JEPA+SIGReg and CLIP, ruling out a simple rotation as the entire cross-modal mismatch.

J.2 Incomplete or invalid experiments

Table 24: Runs deliberately excluded from headline claims. Nothing in this table is silently promoted to a completed result.

Experiment	Status and reason for exclusion
First ViT-S prototype	checkpoint lineage/configuration was corrupt; not a valid model comparison
First spectrum-v2 run	nominal epochs under-consumed rows because of double worker sharding; replaced by corrected 172,390-update run
95k scratch AstroJEPA	useful feasibility pilot, but affected by the older epoch-accounting loader; values are labeled as a pilot
Spectrum continuation pretraining	paused without a usable evaluation checkpoint
No-SIGReg scratch control	not run; collapse prevention is therefore not an empirical ablation in this study
Full pretrained fine-tuning	not run; prevents causal claims about initialization versus scratch
Image local-token JEPA	designed/discussed but not trained
Spectrum-overlap audit	code path exists, but no complete result suitable for reporting
SDSS/VIPERS extension	not run because wavelength grids, masks, resolution, and selection were not yet harmonized

J.3 Artifact-to-claim map

Table 25: Repository experiment families audited for this manuscript.

Artifact family	Paper disposition
Evals/cross_modal_backbone_eval	final 307k probes, 95k pilot, nonlinear probes, checkpoint progression; Sections 3.1, D, E
sequentialvsjoint	central strategy study, all tables and five summary figures; Section 3.2, Appendices E–F
jepavscclip	controlled objective study, all ten figure families and machine-readable tables; Appendix G
galaxy10_checkpoint_evolution	model selection, uncertainty, per-class results; Appendices D, H, I
gzd5_morphology_probe	public split plus protocol-sensitive variants; Appendix H
vit_patch_attention_diagnostics	qualitative attention/PCA, explicitly non-causal; Appendix I
spectra-redshift-diagnostics	SSL-versus-paired distribution shift; Appendix I
checkpoint_loss_curves	training-health diagnostic; summarized, not treated as a separate scientific result

Statistical scope. Probe seeds quantify split/head variation, not representation-training variation. Retrieval and geometry metrics are deterministic over fixed representations. The study has one

378 principal representation-training seed per objective or strategy, so no confidence interval should be
379 interpreted as end-to-end model uncertainty.

380 **Reproducibility.** The repository retains training entry points, frozen representation caches,
381 machine-readable JSON/CSV metrics, split metadata, checkpoint manifests, figure-generation code,
382 and study reports. The paired loaders record source mixture, masks, and uncertainty availability;
383 checkpoints record source paths and training state. Large datasets and checkpoints are not embed-
384 ded in the paper directory.

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Justification: Sections 2–3 give the comparison boundary; Appendices B and C provide splits, augmentations, hyperparameters, and metrics.

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7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Probe error bars quantify split/head variation, but each principal representation has one training seed. This distinction is stated in Sections 3.1, 5, and Appendix J.

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